

Pour Pal v2.0

CMPE 3815 Final Project

Daniel Nalepa, Adam Slamani, Carter Hill



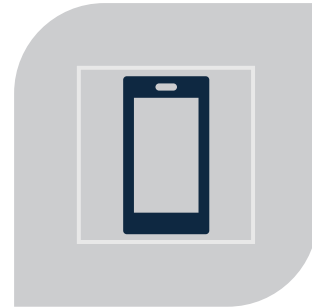
Agenda



HARDWARE
OVERVIEW



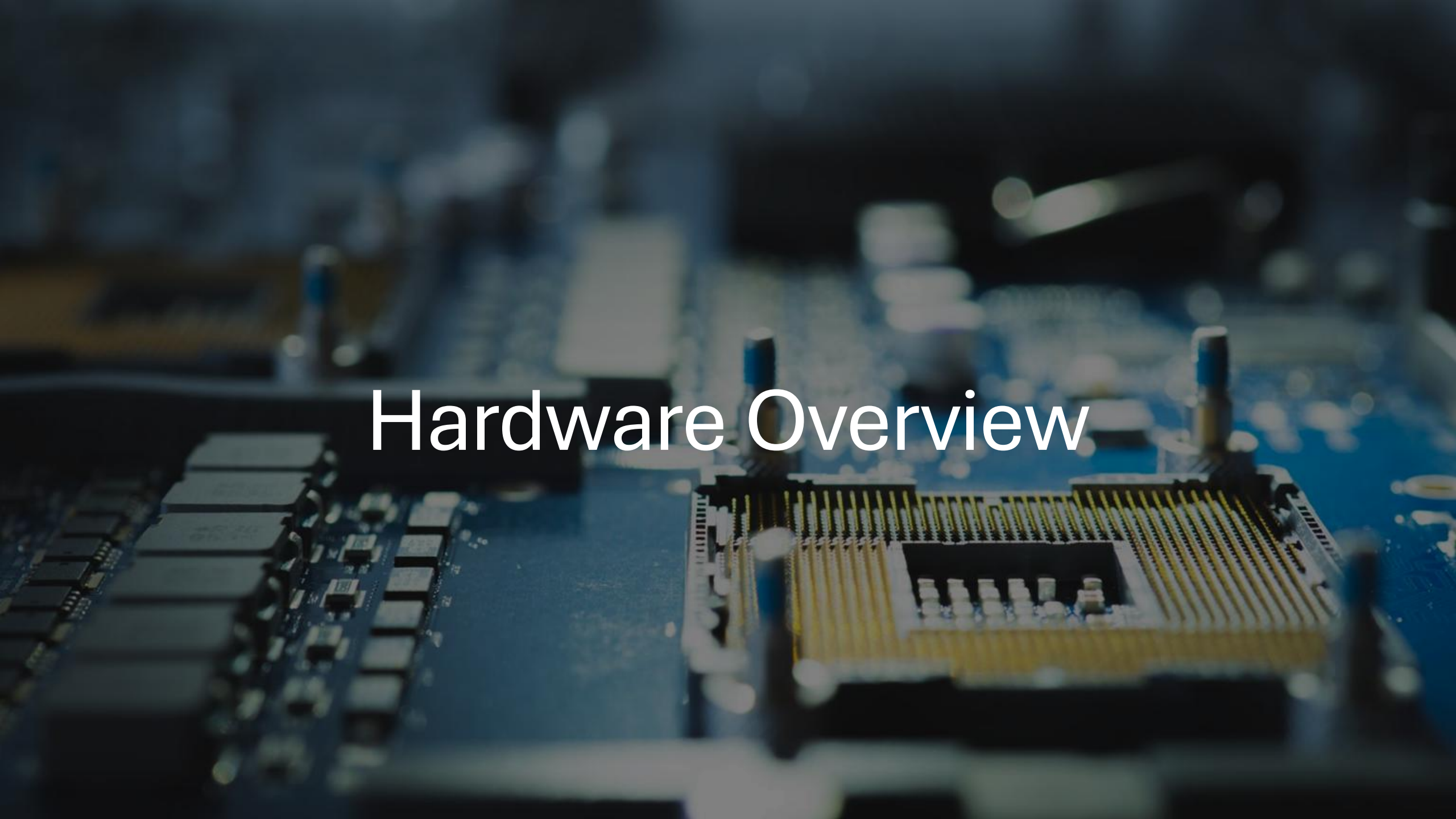
HOUSING
OVERVIEW



CODE OVERVIEW



DEMONSTRATION

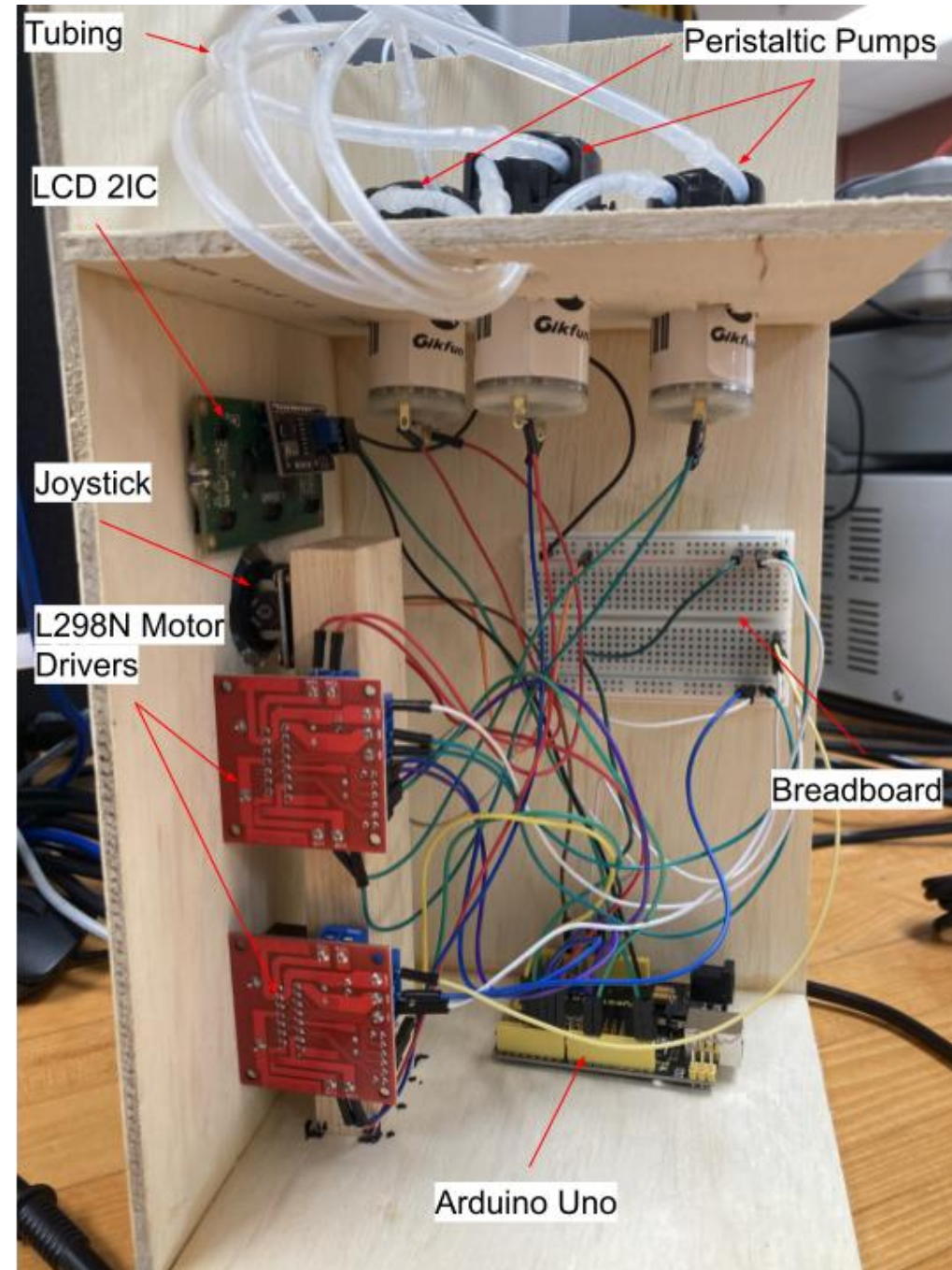


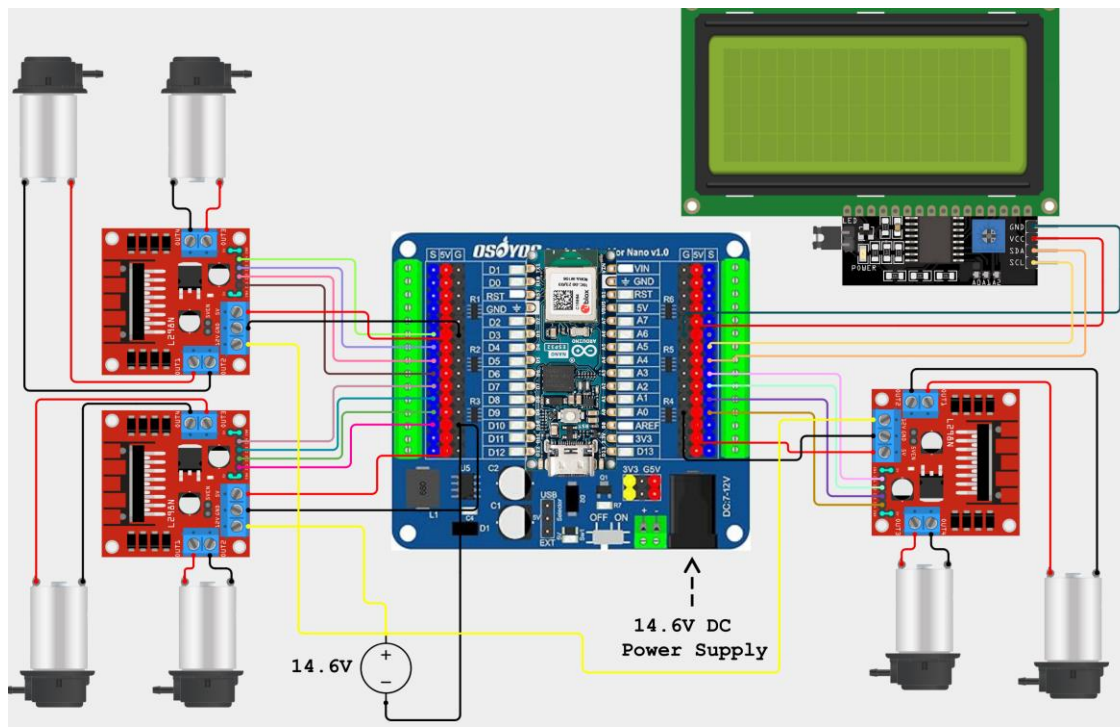
Hardware Overview

Previous Design

Limitations

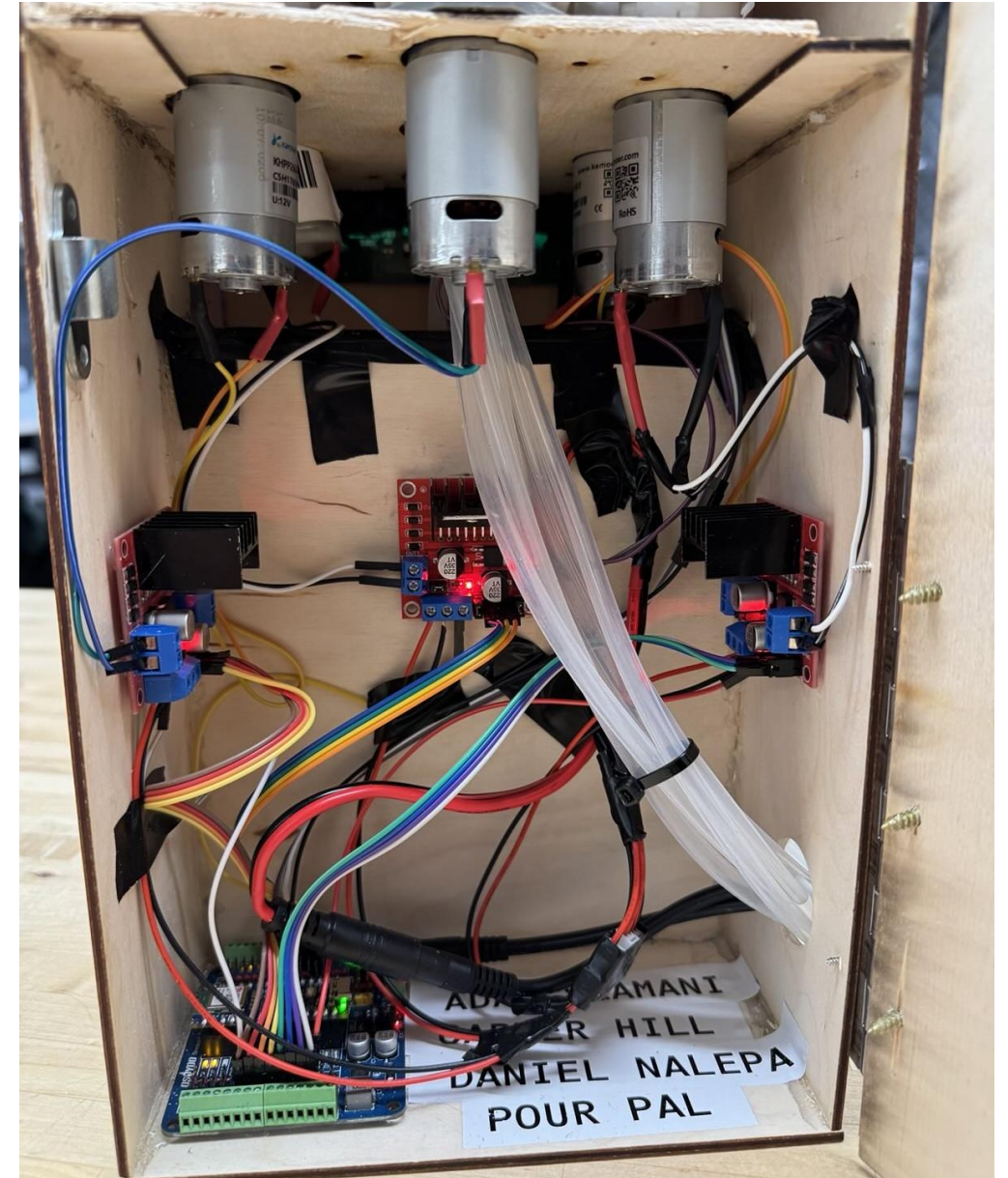
- Only three 100 ml/min pumps
- Motors not at 12VDC so slower than 100 ml
- Joystick controls everything
- Exposed Components
- Ugly housing and prone to spillage





Hardware Overview

- Arduino Nano ESP32
- Breakout Board
- 3 L298N Motor Driver Board
- 6 Peristaltic Pump
- LCD
- 14.6V AC/DC Supply



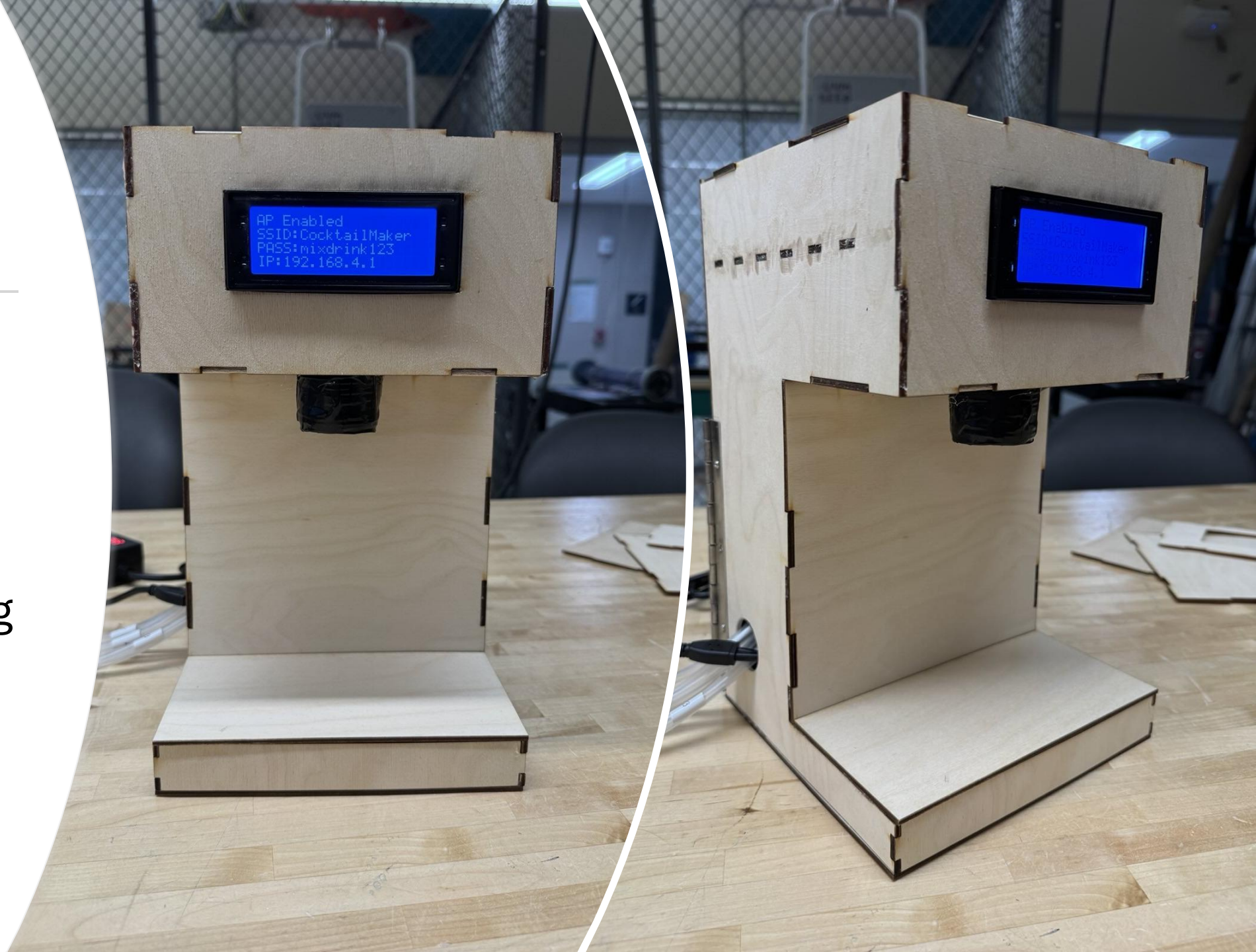
Assembly Design

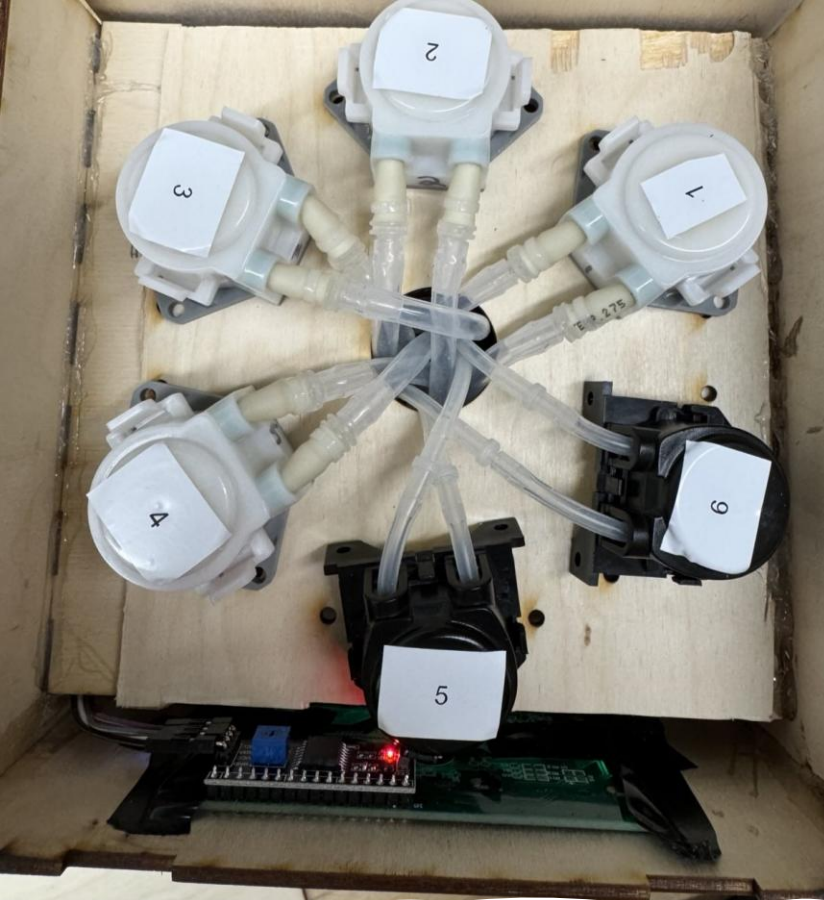


Housing Overview

The Primary Objectives:

- Intuitive Layout
- Easy Assembly
- Troubleshooting Accessibility





More housing pictures

- Internal pump layout and spacing
- Hinged back door with magnetic clasp
- Intake tubing and power cable entrance

Code Overview

Hardware Control

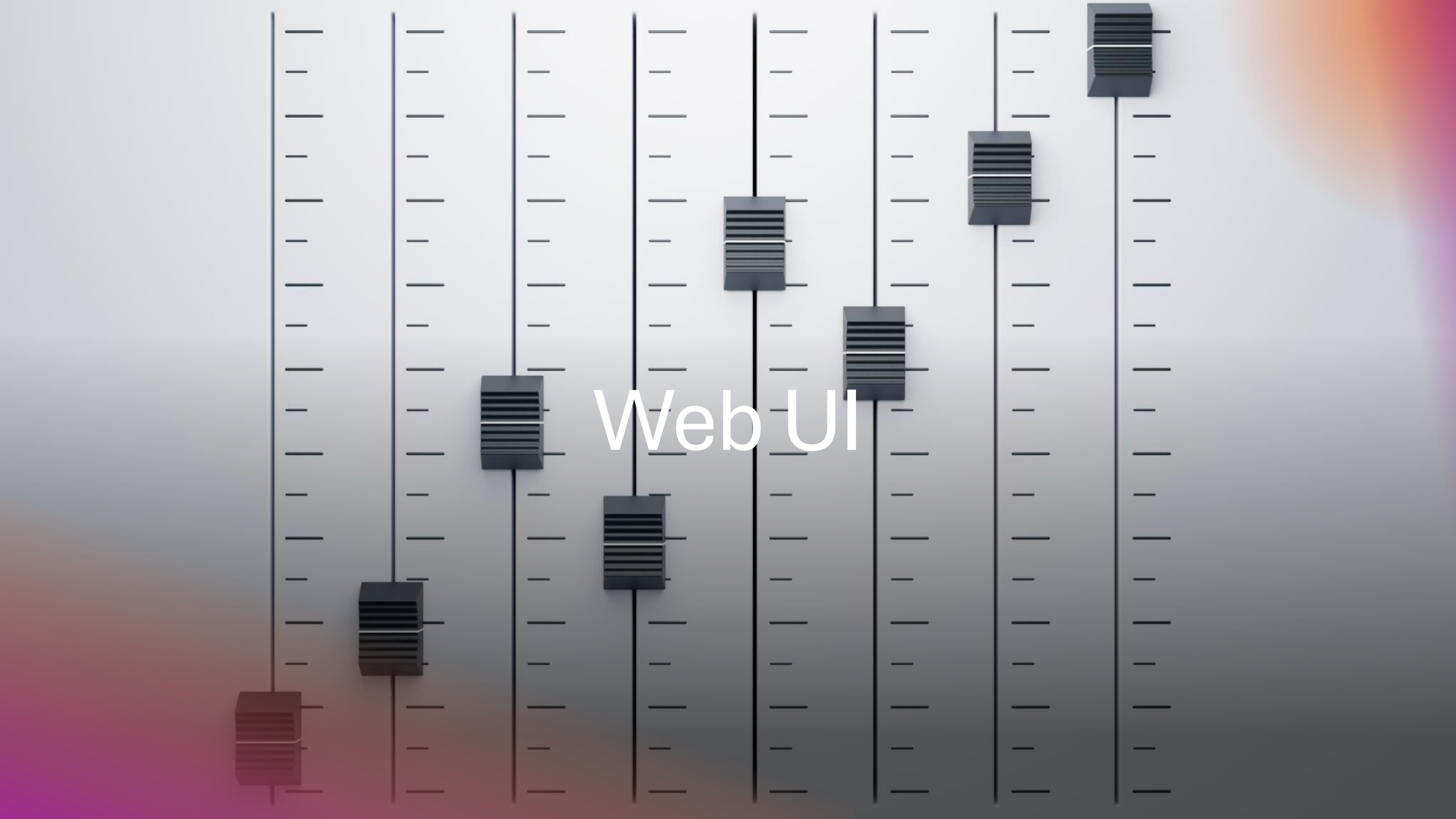
- Helper functions that control motor behavior
- Dispenses by converting from mL→run-time
- Includes ESP32 flash memory with JSON & Preferences library
- Cocktail names (max 10), ingredient name, and pump volumes

```
const float pumpFlowRate[6] = {3.3, 3.3, 3.3, 3.3, 1.6, 1.6};  
void startPump(int idx, int volume_mL) {  
    if(idx<0||idx>=6||volume_mL<=0) return; // Checks for Correct Indexing  
    pumpVolumes[idx] = volume_mL; // Store Requested Volume for said Pump  
    pumpRunning[idx] = true; // Set state of Pump running to true  
    unsigned long duration = (unsigned long)((volume_mL / pumpFlowRate[idx])*1000.0);  
    pumpEndTime[idx] = millis() + duration; // Future Time to Stop Pump  
    hardwareStartPump(idx); // Run Pump
```


Wi-Fi Access Point & Web Server

- The ESP32 creates its own Wi-Fi network.
- A local web server hosts an HTML interface where the user can:
 - ✓ Manually control each pump
 - ✓ Save custom cocktail recipes
 - ✓ Dispense saved cocktails
 - ✓ Clean the tubing
- View system status and uptime
- Uses HTTP Routes that return JSON or perform actions like /dispense, /saveCocktail, and /status

```
// Start Web Endpoints and Server
server.on("/", handleRoot);
server.on("/status", handleStatus);
server.on("/getCocktails", handleGetCocktails);
server.on("/saveCocktail", HTTP_POST, handleSaveCocktail);
server.on("/dispense", handleDispense);
server.on("/dispenseCocktail", handleDispenseCocktail);
server.on("/deleteCocktail", HTTP_POST, handleDeleteCocktail);
server.on("/cleanse", handleCleanse);
server.begin();
Serial.println("Web server started");
```

The background features a series of nine vertical black lines. Each line has a series of horizontal tick marks. Sliders, which are dark grey rectangular blocks with horizontal ridges, are positioned at various heights on these lines. The sliders are arranged in a staggered, ascending pattern from left to right. The first slider on the far left is reddish-brown, while the others are dark grey. The text 'Web UI' is centered in the middle of the image.

Web UI

Not secure192.168.4.1

ESP32 Cocktail Dispenser

Manual Control

Pump 1: 0 mL

Pump 2: 0 mL

Pump 3: 0 mL

Pump 4: 0 mL

Pump 5: 0 mL

Pump 6: 0 mL

Dispense

Cleanse Tubing

Wi-Fi

UVM

Connected

UVMAirLab

CocktailMaker

Secured

Connect automatically

Connect

DIRECT-bb-HP M203 LaserJet

More Wi-Fi settings

Create Custom Cocktail

Cocktail Name:

Pump Liquid Names (1-6)

Pump Volumes (0-100 mL)

Save Cocktail

Stored Cocktails

Daniel
Daniel
Adam
Adam

Dispense Selected

Delete Selected

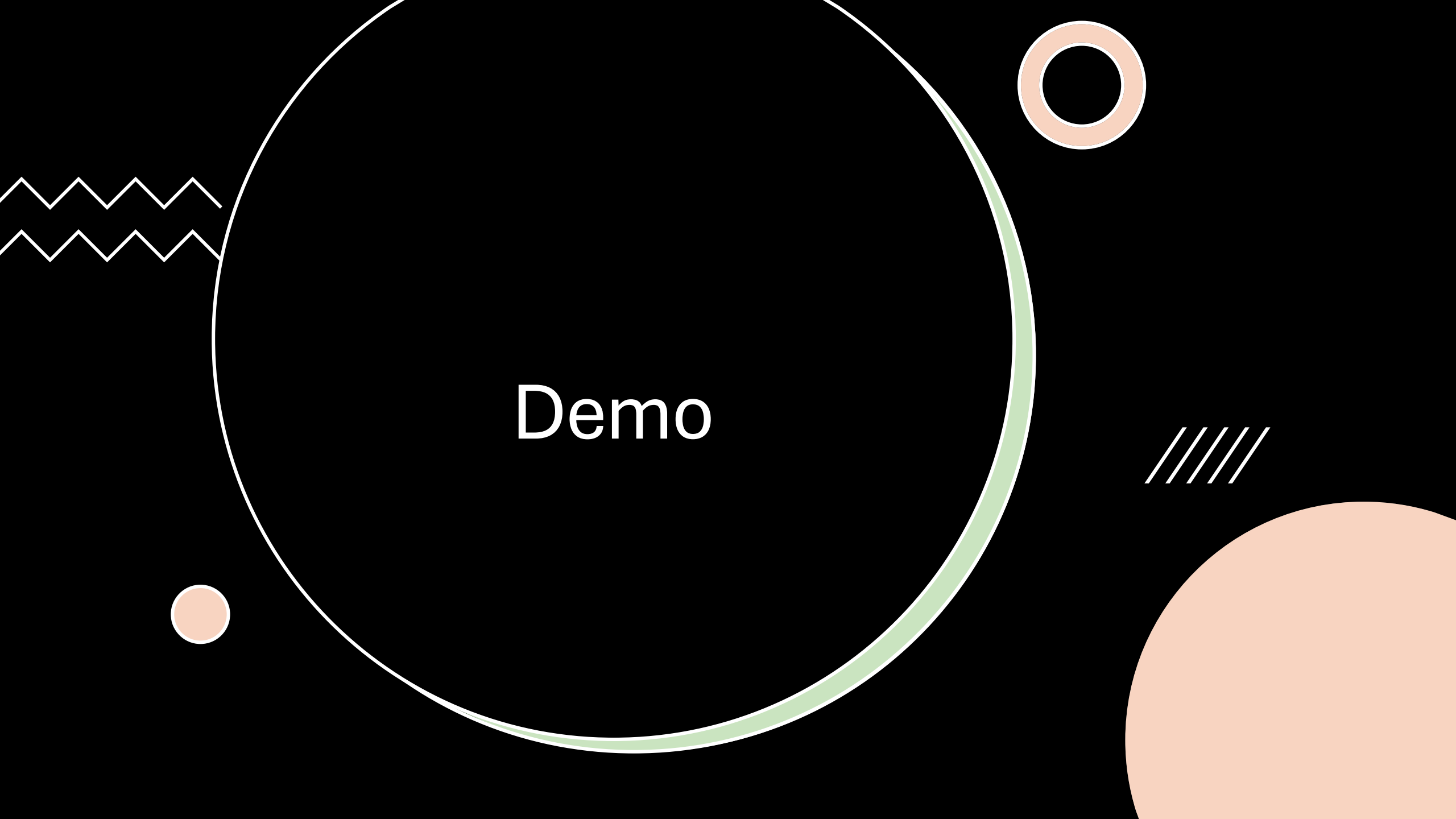
About / Status

IP Address: 192.168.4.1

Uptime: 00:03:40

WiFi: Access Point

- Pumps:
- Pump 1: Idle
 - Pump 2: Idle
 - Pump 3: Idle
 - Pump 4: Idle
 - Pump 5: Idle
 - Pump 6: Idle



The image features a central white circle with the word "Demo" inside. To the left, two white zigzag lines extend from the edge of the circle. In the top right, there is a small orange circle with a white outline. In the bottom left, there is a small orange circle with a white outline. In the bottom right, there is a large orange circle. To the right of the central circle, there are five parallel white diagonal lines. The background is black.

Demo